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AIM202

Use Amazon SageMaker to develop high-quality ML models faster

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Amazon Web Services

Agenda

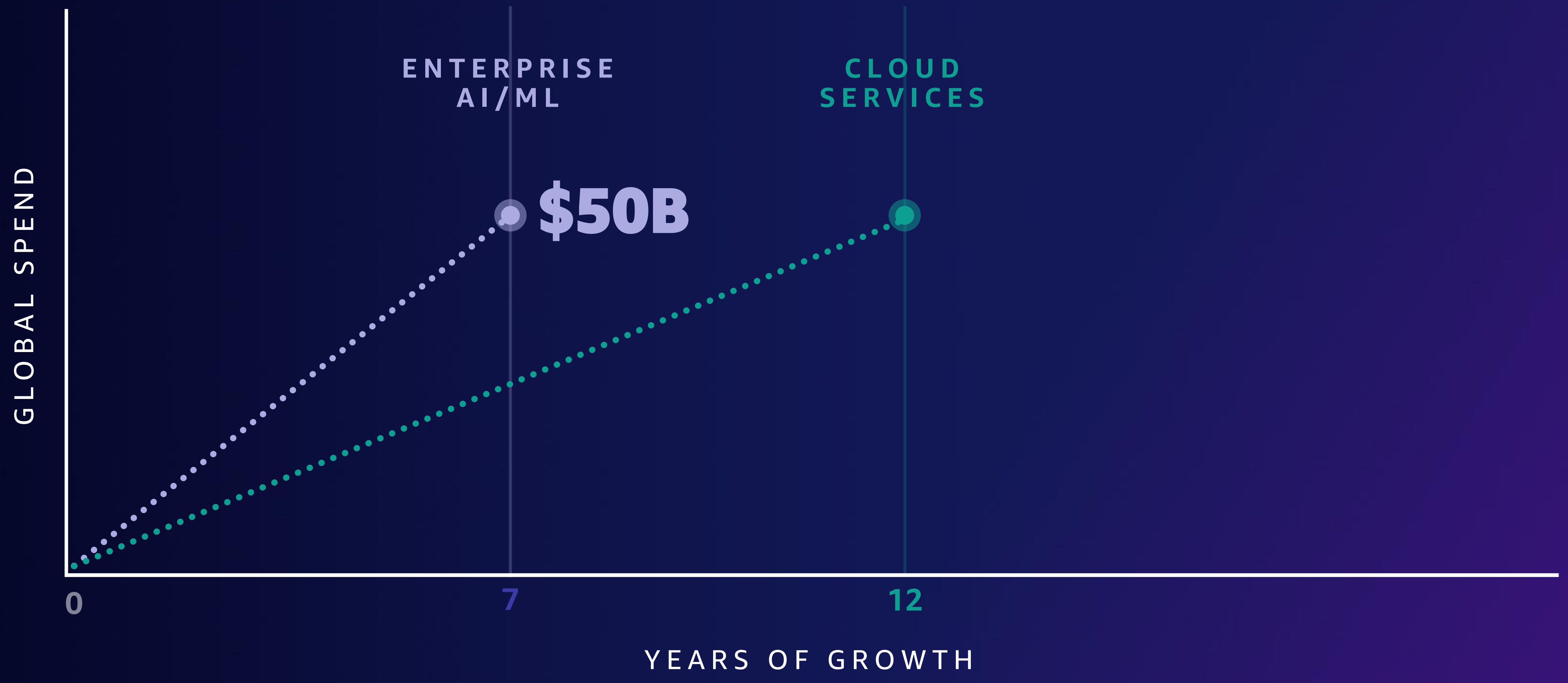
ML overview and personas

Low-code and no-code options to build models

Build, train, deploy on Amazon SageMaker

Q&A

ML is key to innovation



Source: IDC's Worldwide Semiannual Artificial Intelligence Spending Guide, Publication August 2021; IDC Semiannual Public Cloud Services Tracker, 1H2021, November 11, 2021
Note: Enterprise AI/ML and Cloud Services (infrastructure and platform services) categories are not mutually exclusive

How ML drives value creation

Identify risk

Predict churn

Predict new business

Business requirements

Business leads, domain experts, and business analysts

Prioritized use cases →

Data preparation and feature engineering

Data engineers and data scientists

→

Model development, training, and tuning

Data scientists and ML engineers

→

Model deployment and monitoring

ML engineers

Usually takes from weeks to months

primarily solving for the prioritized use cases

Overcoming the barriers to ML

Not enough ML builders	**No-code ML tools** Make ML predictions regardless of ML experience
Processing and labeling massive volumes of data for ML	**Purpose-built data preparation tools** Label and prepare data for ML
Disparate data science tools	**Integrated ML tools in a single interface** Build, train, and deploy models using IDEs
Tedious, manual ML operations	**Built-in MLOps capabilities** Implement MLOps practices to streamline the ML lifecycle

AWS machine learning stack

BROADEST AND MOST COMPLETE SET OF ML CAPABILITIES

AI services

SPECIALIZED

Business processes
Amazon Personalize
Amazon Forecast
Amazon Fraud Detector
Amazon Lookout for Metrics

Search
Amazon Kendra

Code + DevOps
Amazon CodeGuru
Amazon DevOps Guru

Industrial
Amazon Monitron
Amazon Lookout for Equipment
Amazon Lookout for Vision

Healthcare
Amazon HealthLake
Amazon Comprehend Medical
Amazon Transcribe Medical

CORE

Text/documents
Amazon Textract
Amazon Comprehend
Amazon Translate

Chatbots
Amazon Lex

Speech
Amazon Polly
Amazon Transcribe
Amazon Transcribe Call Analytics

Vision
Amazon Rekognition
AWS Panorama

ML services

Label data

SAGEMAKER CANVAS
ML for business analysts

SAGEMAKER STUDIO LAB
Learn ML

AMAZON SAGEMAKER STUDIO IDE
Prepare data | Store features | Detect bias | Build with notebooks | Train models | Tune parameters | Deploy in production | Explain predictions | Manage & monitor

Manage edge devices

ML frameworks & infrastructure

TensorFlow, PyTorch, Apache MXNet, Hugging Face

Deep learning AMIs & containers

Amazon EC2 | CPUs | GPUs | AWS Inferentia | AWS Trainium | Habana Gaudi | FPGA

SageMaker

SageMaker helps organizations harness ML

Business analysts

Make ML predictions using a visual interface with Amazon SageMaker Canvas

SageMaker

Infrastructure, tools, visual interfaces, workflows, orchestration, and collaboration

Data scientists

Prepare data and build, train, and deploy ML models with Amazon SageMaker Studio

MLOps engineers

Deploy and manage models at scale with Amazon SageMaker MLOps

Tens of thousands of customers use SageMaker

3M Science. Applied to Life.
AGCO
amazon
Atlas Van Lines
AUTODESK
avis budget group
BUNDESLIGA
Celgene
Capital One
Cerner
CONDÉ NAST
CONVOY
Domino's
FICO
Formosa Plastics
Frame.io
GoDaddy
lyft
HEARST
ifood
Infoblox NEXT LEVEL NETWORKING
intuit
INVISTA
Lenovo
Liberty Mutual INSURANCE
NFL
NOVARTIS
Roche
NuData Security mastercard
THOMSON REUTERS
tinder
verizon
arlo
coinbase
freshworks
INTERCOM
grammarly
SIEMENS
EULER HERMES
T Mobile
BAYER
mercado libre
GP Georgia-Pacific

aws

SageMaker

Common use cases

Predictive maintenance
Manufacturing, automotive, IoT

Demand forecasting
Retail, consumer goods, manufacturing

Fraud detection
Financial services, online retail

Credit risk prediction
Financial services, retail

Extract and analyze data from documents
Healthcare, legal, media/ent, education

Computer vision
Healthcare, pharma, manufacturing

Autonomous driving
Automotive, transportation

Personalized recommendations
Media and entertainment, retail, education

Churn prediction
Retail, education, software and internet

SageMaker no-code and low-code ML tools

SageMaker Canvas

ML requires a lot more builders

2x

the growth of any other emerging job role

+74%

annual growth in the past 4 years

Source: LinkedIn Emerging Jobs Report 2020

Amazon SageMaker Canvas

Generate ML predictions – no code required

- Quickly access and prepare data sources for ML
- AutoML built in to generate accurate predictions
- Share ML models with data science teams

SageMaker Overview

SageMaker

PREPARE

SageMaker Ground Truth
Label training data for machine learning

SageMaker Data Wrangler
Aggregate and prepare data for machine learning

SageMaker Processing
Built-in Python, BYO R/Spark

SageMaker Feature Store
Store, update, retrieve, and share features

SageMaker Clarify
Detect bias and understand model predictions

BUILD

SageMaker Studio Notebooks
Jupyter notebooks with elastic compute and sharing

Built-in and bring-your-own algorithms
Dozens of optimized algorithms or bring your own

Local mode
Test and prototype on your local machine

SageMaker Autopilot
Automatically create machine learning models with full visibility

SageMaker JumpStart
Pre-built solutions for common use cases

TRAIN & TUNE

One-click training
Distributed infrastructure management

SageMaker Experiments
Capture, organize, and compare every step

Automatic model tuning
Hyperparameter optimization

Distributed training
Training for large datasets and models

SageMaker Debugger
Debug and profile training runs

Managed spot training
Reduce training cost by 90%

DEPLOY & MANAGE

Fully managed deployment
Fully managed, ultra-low latency, high throughput

Kubernetes & Kubeflow integration
Simplify Kubernetes-based machine learning

Multi-model endpoints
Reduce cost by hosting multiple models per instance

SageMaker Model Monitor
Maintain accuracy of deployed models

SageMaker Edge Manager
Manage and monitor models on edge devices

SageMaker Pipelines
Workflow orchestration and automation

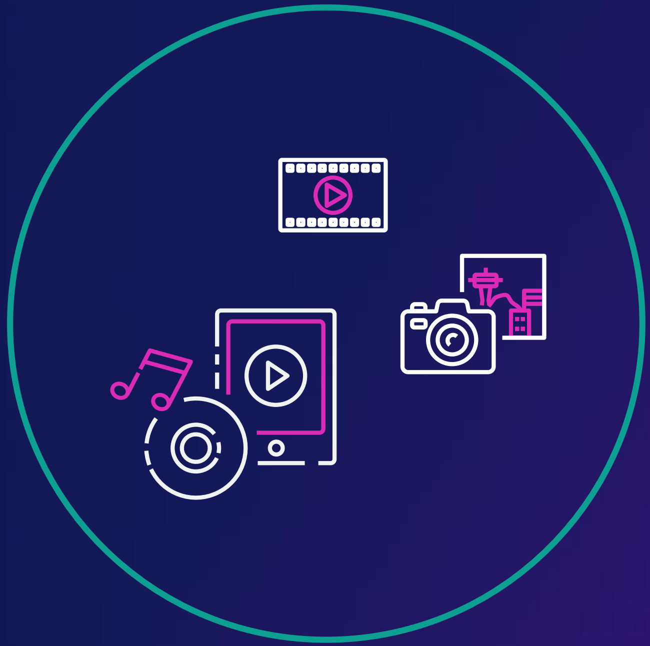
SageMaker Studio
Integrated development environment (IDE) for ML

Not a comprehensive list. Visit https://aws.amazon.com/sagemaker for the latest information

Purpose-built data preparation tools

SageMaker Data Wrangler, SageMaker Feature Store, and SageMaker Ground Truth Plus

Structured



Unstructured

Amazon SageMaker Data Wrangler

No-code data preparation

- Single visual interface for common data prep techniques
- Select data from multiple sources
- 300+ built-in transformations to prepare data without writing code

Amazon SageMaker Feature Store

Securely store, discover, and share features for ML

- Online and offline
- Millisecond latency
- Consistent features
- Visual search
- Sharing and collaboration

Unstructured data = big opportunity, but underutilized

Structured

Amazon SageMaker Ground Truth Plus

Deliver high-quality training datasets fast and reduce data labeling costs

- Increase data quality through ML-powered data labeling
- Access expert data labelers
- Reduce data labeling costs with assistive labeling features
- Improve operational efficiency by reviewing project metrics
- Make data labeling accessible to data operations and program managers

PROBLEM

Wanted to speed up all stages of research and development, particularly the review of tissue samples

Labeling data was time-consuming and tedious

SOLUTION

Worked with the Amazon ML Solutions Lab to integrate SageMaker Ground Truth to automate tedious portions of work

Helps annotate, collect, and classify ML model training data quickly

IMPACT

Reduces time humans spend cataloging samples by at least 50%

Accelerates the drug research process and the introduction of medicines to the market

Increases the pace of research by freeing up scientists

ASTRAZENECA PLC

AstraZeneca

Integrated ML tools in a single interface

SageMaker Studio, SageMaker JumpStart, SageMaker AutoPilot

Amazon SageMaker Studio

Fully integrated development environment (IDE) for ML

Increased productivity

Code, build, train, deploy, and monitor in a unified visual interface

Collaboration at scale

Share notebooks without tracking code dependencies

Visualize experiment management

Visualize how you organize, track, and compare thousands of experiments

Higher-quality ML models

Automatically debug errors, monitor models, and maintain high quality

SageMaker Studio integrated capabilities

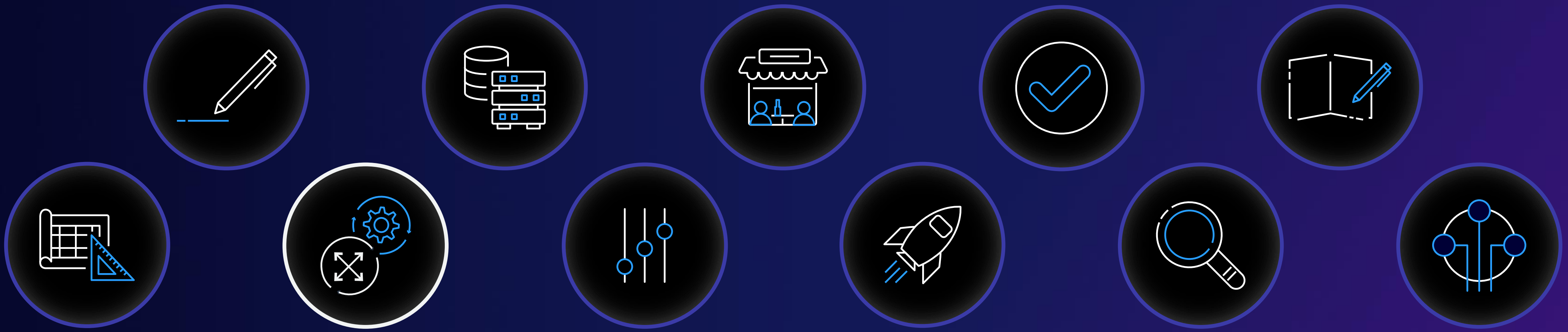
Label data

Collect and prepare data

Store, search, use features

Check data

Visualize in notebooks



Pick algorithm

Train models

Tune parameters

Deploy in production

Manage and monitor

CI/CD

SageMaker Studio IDE

Build

Train

Deploy

MLOps

on SageMaker

Build ML models

Fully managed, shareable notebooks on Amazon EC2

Fully managed, shareable Jupyter notebooks
Run notebooks on elastic compute resources

Built-in algorithms
15 built-in algorithms available in prebuilt container images

Prebuilt solutions and open-source models
Over 150 popular open-source models

AutoML
Automatically create ML models with full visibility

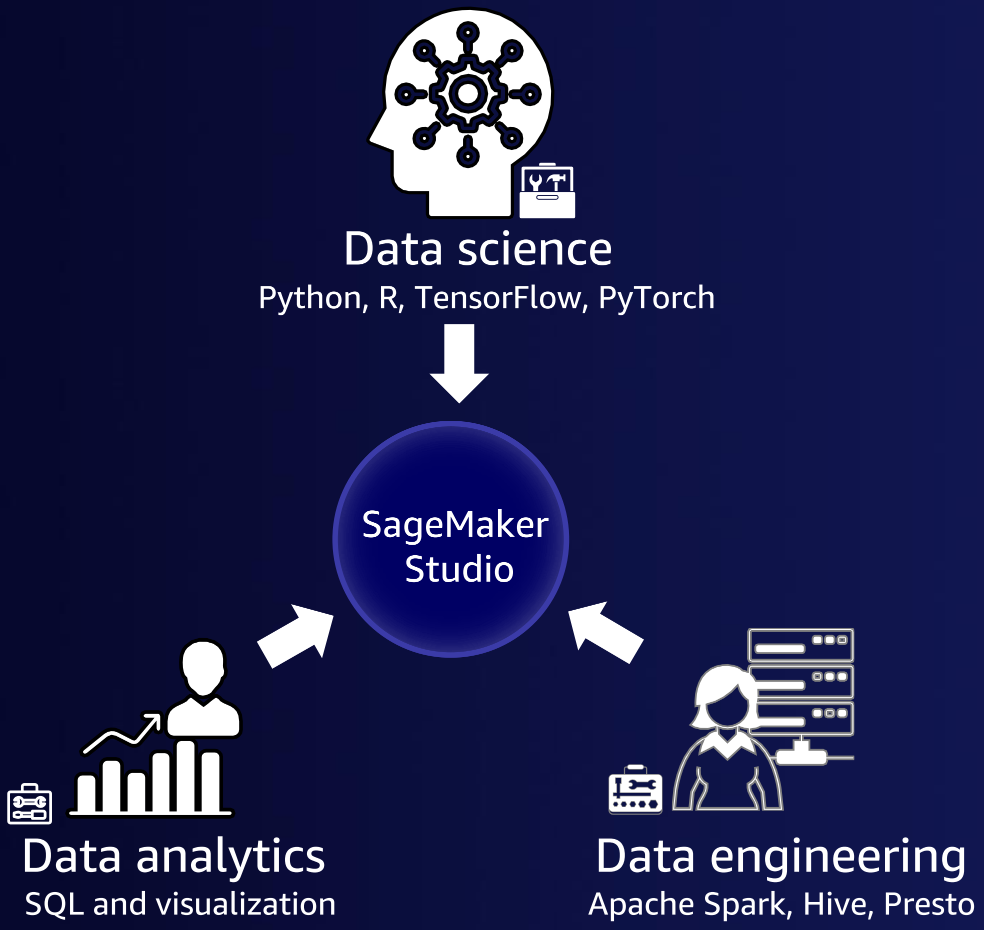
Support for major frameworks and toolkits
Optimized for popular deep learning (DL) frameworks such as TensorFlow, PyTorch, Apache MXNet, and Hugging Face

Amazon SageMaker Studio notebook

Perform data engineering, analytics, and ML workflows in one notebook

- **Connect with Amazon EMR, Amazon S3, and more**
- **Interactively access, transform, and analyze a wide range of data**
- **Build, train, and deploy models using your preferred framework**

Universal notebook for data preparation and ML



Drivers

- Data preparation and analytics are foundational components of ML workflows
- Switching between multiple notebooks, tools, and interfaces reduces productivity
- Security and access control need to be consistent across analytics and ML services

Interactive ML with SageMaker Studio notebooks

- Comprehensive data preparation and ML from the same notebook
- Discover, create, and manage Amazon EMR clusters from SageMaker Studio across accounts
- One-click monitoring and debugging Spark jobs from SageMaker Studio

PROBLEM

3 TB+ data, 1,500+ hours play time per week

Needed ML solution for real-time stats

Lean team, no data science expertise

SOLUTION

Next Gen Stats (NGS):

Live data streamed to AWS from RFID tags on players and embedded in game ball

Data processed in 100+ steps in under 1 second

Real-time predictions

Output stats published through APIs and on-screen graphics

IMPACT

Quickly launched 20+ stats

Sports announcers get interesting data points to engage fans

PROBLEM

Ingests thousands of data points per day from the American secondary mortgage market

Wanted to leverage data more efficiently by implementing ML into its tools

SOLUTION

Used SageMaker to create ML models that help evaluate everything from loan risk assessment to property value analysis for its multifamily line of business

IMPACT

Helped the company improve its ability to identify applicants who are likely to default from 3.5% to 48%

Provides transparency in the home loan process for lenders, investors, and analysts

THE FEDERAL NATIONAL MORTGAGE ASSOCIATION (FANNIE MAE)

Fannie Mae®

Learn and experiment with machine learning

Amazon SageMaker Studio Lab (Preview)

Free, no-configuration development environment

All you need is an email address to get started

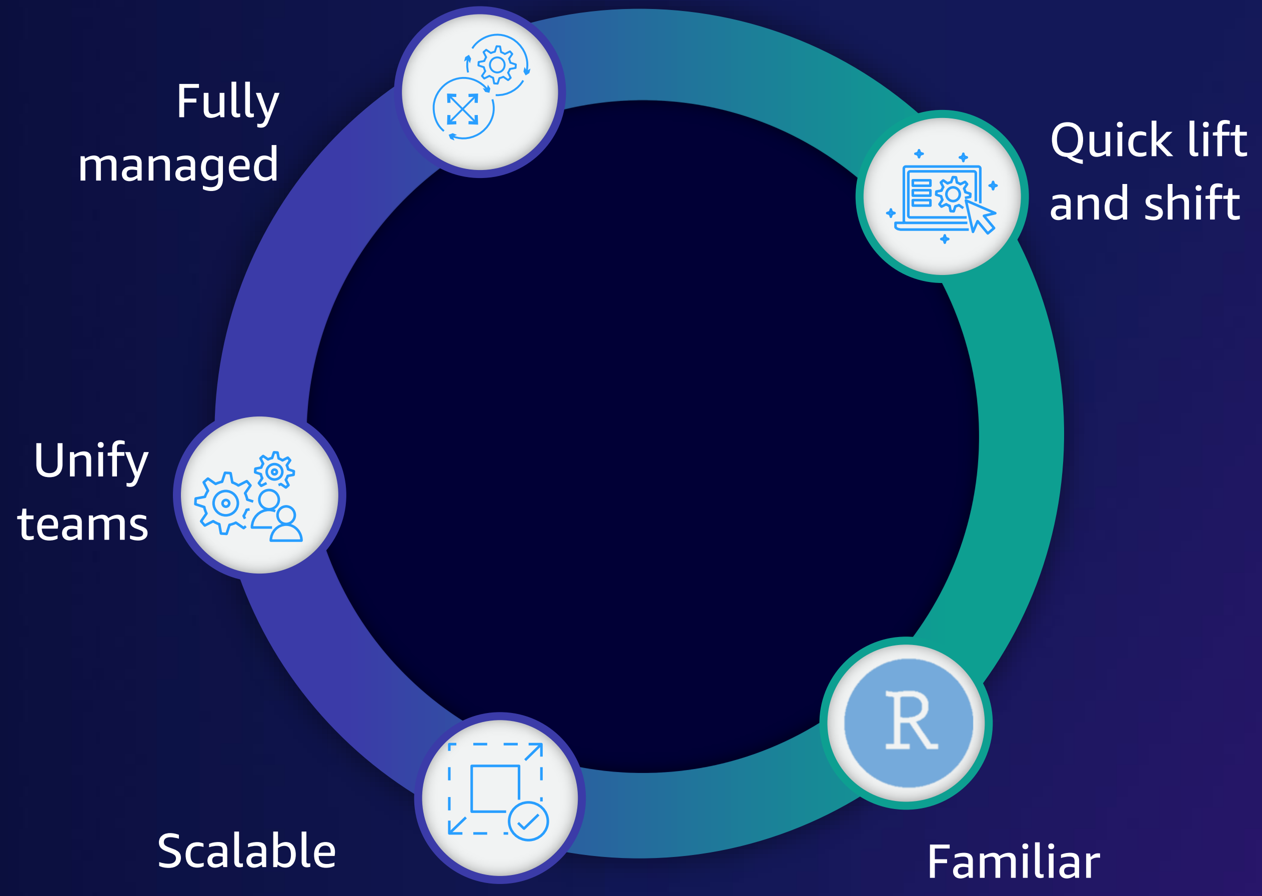
As many CPU and GPU sessions as needed

Automatically save session data

Models ready for production

RStudio on SageMaker

USE RSTUDIO AND SAGEMAKER STUDIO FOR DATA SCIENCE



Agility with a unified data science

Python developers

Use SageMaker's IDE to

Explore data

Build models

Deploy and manage models in production

R developers

Use RStudio's IDE to

Perform data analysis

Build statistical models

Process data

Build

Train

Deploy

MLOps

on SageMaker

Train ML models

Fast and cost-effective ML model training

Experiment management and model tuning
Save weeks of effort by automatically tracking training runs and tuning hyperparameters

Debug and profile training runs
Use real-time metrics to correct performance problems

Distributed training
Complete distributed training up to 40% faster

Training compiler
Accelerate training times by up to 50% through more efficient use of GPUs

Managed spot training
Reduce the costs of training by up to 90%

PROBLEM

Develops computer vision models for autonomous driving

Needed to reduce training times to accelerate model development

SOLUTION

SageMaker data parallelism library to speed up training jobs

Amazon SageMaker Debugger to monitor, trace, and analyze performance issues

Performance optimization with Amazon ML Solutions Lab

IMPACT

10x faster model training using SageMaker data parallelism library and ML Solutions Lab's code optimization

Hyundai Motor Company can now focus on data preparation and accelerate the development cycle

HYUNDAI MOTOR COMPANY

Build

Train

Deploy

MLOps

on SageMaker

Deploy ML models

Fully managed deployment for inference at scale

Wide selection of infrastructure
70+ instance types with varying levels of compute and memory to meet the needs of every use case

Single-digit millisecond overhead latency
For use cases requiring real-time responses

Asynchronous inference
Supports large models with long-running processing times

Cost-effective deployment
Multi-model/multi-container endpoints, serverless inference, and elastic scaling

Built-in integration for MLOps
ML workflows, CI/CD, lineage tracking, and catalog

Automatic deployment recommendations
Optimal instance type/count and container parameters, and fully managed load testing

Support the responsible use of ML throughout the model lifecycle using SageMaker Clarify

Build

Perform bias analysis during exploratory data analysis

Train

Conduct bias and explainability analysis after training

Deploy

Explain individual inferences from models in production

Monitor

Validate bias and relative feature importance over time

Build

Train

Deploy

MLOps

on SageMaker

Built-in MLOps capabilities

SageMaker MLOps, SageMaker Pipelines

Operational challenges with managing the ML lifecycle

Manual iterative processes slow down ML innovation

Difficult to scale the number of models in production

CI/CD for ML requires writing custom code

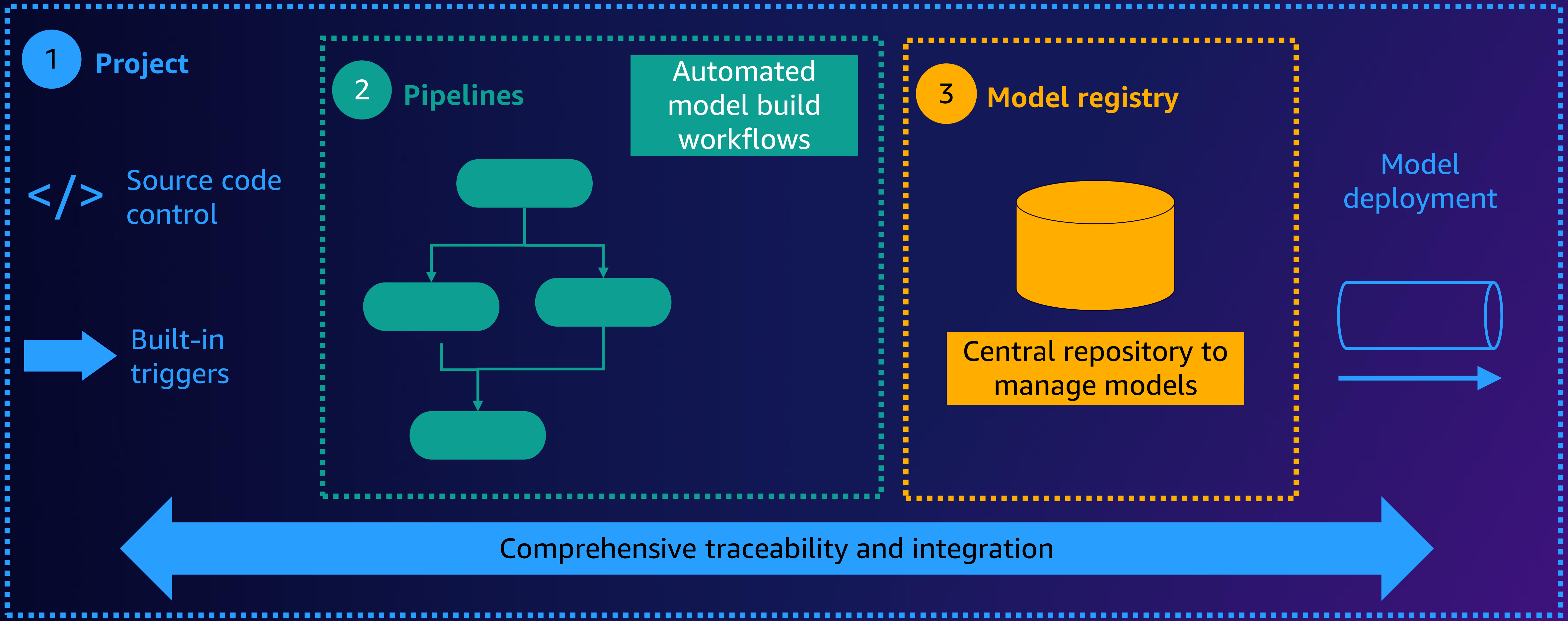
Compliance requirements are difficult to meet

Amazon SageMaker MLOps

Streamline the ML lifecycle

- **Automate ML workflows to scale model development**
- **Build CI/CD pipelines for ML to accelerate model deployment**
- **Catalog model versions, metadata, metrics, and approvals for traceability and reusability**
- **Track lineage for troubleshooting and compliance**
- **Maintain accuracy of predictions after models are deployed**
- **Enhance governance and security**

SageMaker – Components



SageMaker benefits and next steps

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SageMaker key benefits

Most complete, comprehensive ML service

Democratize ML innovation
Empower more groups of people, including business analysts

Prepare data at scale
Access, label, and process structured and unstructured data

Accelerate the ML lifecycle
Reduce training time from hours to minutes

Streamline ML processes
Automate and standardize MLOps processes

SageMaker

GETTING STARTED

Business analysts

Get started with SageMaker Canvas

No-code ML

Data scientists

Onboard through SageMaker Studio

Single IDE for full-code ML

MLOps engineers

Implement MLOps with SageMaker Pipelines

Workflow automation and CI/CD

Learn in-demand AWS Cloud skills

AWS Skill Builder

Access **500+ free** digital courses and Learning Plans

Explore resources with a variety of skill levels and **16+** languages to meet your learning needs

Deepen your skills with digital learning on demand

Train now

AWS Certifications

Earn an industry-recognized credential

Receive Foundational, Associate, Professional, and Specialty certifications

Join the **AWS Certified community** and get exclusive benefits

Access **new** exam guides

Thank you!